Artosphere (ARTS) Whitepaper

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Table of Contents

1. Abstract
2. Scientific Foundation
3. Protocol Architecture
4. Tokenomics
5. Governance
6. Experimental Roadmap
7. Risk Disclosure
8. Deployed Contracts
9. References

1. Abstract

Artosphere (ARTS) is a decentralized science (DeSci) protocol on Base mainnet whose token emission is triggered by real-world physics experiments. When an experiment publishes results relevant to an on-chain prediction, the protocol mints a predetermined amount of ARTS and distributes it automatically across a grant fund, liquidity pool, staking rewards, and burn.

The underlying scientific hypothesis proposes that certain Standard Model parameters obey discrete algebraic identities over the ring $\mathbb{Z}[\varphi] = \mathbb{Q}(\sqrt{5})$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. One Tier 1 prediction — the solar neutrino mixing angle $\sin^2\theta_{12} = 1/(2\varphi) \approx 0.30902$ — has been experimentally confirmed by the JUNO collaboration at 0.021σ. Five additional decisive experimental tests are scheduled between 2026 and 2045.

The maximum token supply is 987,000,000 ARTS = $F(16) \times 10^6$. Supply is science-gated: new ARTS can only be minted through the Experiment Milestone Protocol upon sector resolution, which requires either AI oracle consensus or human arbitration by ORCID-verified researchers.

2. Scientific Foundation

2.1 The Hypothesis

The Artosphere Hypothesis is a falsifiable proposal that certain Standard Model parameters obey discrete algebraic identities over the ring $\mathbb{Z}[\varphi] = \mathbb{Q}(\sqrt{5})$, with a single dimensional constant (M_Planck) and a single dimensionless constant ($\varphi = (1+\sqrt{5})/2$).

The mathematical structure has a category-theoretic interpretation via A_5 (the icosahedral group, PSL(2,5)) and Fibonacci Modular Tensor Category. The scope is algebraic relations within the Standard Model flavor sector; the hypothesis does not

propose new particles beyond the Standard Model, modified gravity, or a Theory of Everything.

Full derivations and mathematical proofs are published in peer-reviewable depositions on Zenodo (see Section 9). This whitepaper summarizes the results directly relevant to token mechanics.

2.2 Tier 1 Predictions

#	PREDICTION	FORMULA	EXPERIMENTAL STATUS
1	Solar neutrino mixing	$\sin^2\theta_{12} = 1/(2\varphi) = 0.30902$	⭐
2	Strong coupling	$\alpha_s(M_Z) = 1/(2\varphi^3) = 0.11803$	PDG 2024: 0.1180 ± 0.0009 → 0.15σ
3	Leptonic CP phase	$\delta_CP = \pi + \arctan(\sqrt{5}) \approx 243.4°$	T2K/NOvA: consistent within current precision
4	Strong CP	$\theta_QCD = 0$ (topological)	nEDM bound: consistent
5	Critical line identity	$½ - 1/\varphi^2 = \alpha_s$	Algebraic consequence of #1 and #2

— Higgs self-coupling λ_H = (1/2)φ^(−(5+√5)/4), predicting m_H = 125.256 GeV (matches PDG at 0.04σ). The exponent (5+√5)/4 equals D^2(Fib)/2, where D^2(Fib) is the global quantum dimension squared of Fibonacci Modular Tensor Category — a category-theoretic invariant defined a priori, independent of the Higgs sector.

2.3 The JUNO Confirmation

The JUNO (Jiangmen Underground Neutrino Observatory) collaboration released its first neutrino oscillation result in late 2025: , based on

59 days of reactor-antineutrino data. The Artosphere prediction $1/(2\varphi) = 0.309017$ differs from this measurement by 0.021σ.

The prediction was pre-registered in Zenodo depositions in early 2026, before the JUNO measurement was published.

The value $1/(2\varphi)$ (equivalently, $\tan \theta_{12} = 5^{(1/4)}$) is the only "golden ratio" neutrino mixing pattern compatible with current global fit data (NuFIT 5.2) within 1σ. Two prior proposals in the literature — GR1 (Kajiyama, Raidal, Strumia 2007, $\sin^2\theta_{12} = 0.276$) and GR2 (Rodejohann 2009, $\sin^2\theta_{12} = 0.345$) — are now excluded at 3.8σ and 4.1σ respectively. The Artosphere pattern is not found in the published A_5 flavor model literature (Albright–Dueck–Rodejohann 2010 canonical classification), making it an original prediction.

Final JUNO precision (~0.5%) is expected between 2028 and 2030, providing approximately 16σ discrimination between the Artosphere prediction and all competing golden-ratio or tri-bimaximal patterns.

2.4 Honest Probability

The prior probability that the Artosphere Hypothesis describes real physics, as assessed after extensive internal audit:

- Optimistic: 24%
-
- Pessimistic: 8%

The protocol is designed to function and create value across the full range of outcomes, including refutation of one or more Tier 1 predictions.

3. Protocol Architecture

The Artosphere protocol on Base mainnet consists of a token and staking layer (existing) plus an Experiment Milestone layer that translates real-world physics results into on-chain events.

3.1 Token and Staking Layer

- — ERC-20 token with UUPS upgradeability and a 0.618% spiral burn applied to transfers (with exempt addresses for staking and liquidity infrastructure).
- — Lock-tiered staking contract with Fibonacci lock durations (5, 21, 55 days) and golden-ratio tier multipliers (1×, φ, φ^2). Initial APY of 61.8% with weekly epoch decay by a factor of 1/φ.
- — Constant-product automated market maker (spot swaps, no leverage) with a 1.18% swap fee routed to the protocol treasury.
- — Stake-on-prediction contract allowing users to back or oppose specific experimental outcomes.
- — On-chain registry mapping wallet addresses to ORCID-verified researcher identities.

3.2 Experiment Milestone Protocol

The Experiment Milestone Protocol (EMP) is the core innovation of Artosphere. It consists of four smart contracts that collectively translate experimental publications into on-chain state transitions and token emission.

3.2.1 ExperimentMilestoneProtocol.sol

Tracks six experiment sectors. Each sector represents a specific falsifiable prediction tied to a specific experiment and has:

- Name, textual prediction, target experiment
- Predicted numerical value with sigma threshold
- Fixed ARTS emission amount (triggered on resolution)

- State machine: `PENDING → DATA_RECEIVED → CONFIRMED / REFUTED / INCONCLUSIVE / PREDICTION_RECLASSIFIED / SCOPE_REFINED`

a sector transitioning to `CONFIRMED` mints the same amount of ARTS as a sector transitioning to `REFUTED`, with the same distribution. The protocol treats all experimental results as value-creating events.

Sector transitions are permissioned (`ORACLE_ROLE`, held by the ScienceOracle contract). Each transition triggers `MilestoneEmission.onSectorResolved()`.

3.2.2 ScienceOracle.sol

A four-layer oracle system for translating off-chain experimental results into on-chain state.

Daily scraping of arXiv (hep-ph, hep-ex, nucl-ex), PDG API, HEPData, and INSPIRE-HEP. Automated parsing for publications matching each sector's keywords and experimental identifiers.

Two independent large language model agents (Claude and GPT) each extract numerical results from the parsed publication and compute the sigma deviation from the on-chain prediction. Only results where both agents agree advance to on-chain submission.

`submitResult` An authorized submitter (Chainlink Functions node calling from the off-chain pipeline) submits the result on-chain with: sector ID, measured value and sigma deviation, proposed outcome, evidence hash (arXiv DOI or IPFS), and dual-AI consensus proof. This begins a 7-day challenge window. Any address may challenge the submission by posting a 0.001 ETH bond and providing counter-evidence.

- → any address may call `finalizeSubmission()` (permissionless). The result pushes to ExperimentMilestoneProtocol and the sector transitions. - → `ARBITER_ROLE` (ResearcherRegistry 2-of-3 multisig of ORCID-verified physicists) resolves the challenge. If upheld, original submission stands and the challenger forfeits the bond. If overturned, the submission is discarded; a new submission may be made.

3.2.3 MilestoneEmission.sol

Receives callbacks from ExperimentMilestoneProtocol when a sector is resolved. Mints the sector's predetermined ARTS amount via PhiCoin's `mintTo` (requires MINTER_ROLE on PhiCoin) and distributes in fixed proportions:

DESTINATION	SHARE
HypothesisEvolutionFund	50%
Liquidity Pool (Aerodrome)	30%
Staking Rewards Pool	15%
Burn	5%

A sector can only emit once. The contract tracks `sectorEmitted[sectorId]` and reverts on double-emission attempts.

3.2.4 HypothesisEvolutionFund.sol

A grant and bounty contract that receives 50% of every milestone emission and distributes through two channels:

Any address may propose a grant by calling `proposeGrant(recipient, amount, title, ipfsHash, sectorId)`. `APPROVER_ROLE` reviews and either approves (with a note) or rejects. Approved grants are disbursed by admin.

Admin creates bounties with `createBounty(reward, title, ipfsHash, sectorId)`. Any address may submit work via `submitBounty(bountyId, submissionHash)`. APPROVER_ROLE evaluates and releases payment.

The fund is initially seeded with 8,000,000 ARTS and refills through ongoing milestone emissions, creating a self-sustaining research budget that grows with experimental progress.

4. Tokenomics

4.1 Supply

- 987,000,000 ARTS = F(16) × 10^6
- 8,000,000 ARTS (in HypothesisEvolutionFund)
- 979,000,000 ARTS

The unminted supply is science-gated: it can only be minted through the MilestoneEmission contract, which can only be called by ExperimentMilestoneProtocol after a sector is resolved, which requires oracle consensus or arbitration. There is no administrative path to mint ARTS without a real experimental result.

4.2 Milestone Emission Schedule

SECTOR	TARGET EXPERIMENT	EXPECTED RESOLUTION	EMISSION	% OF MAX
STRONG_CP	n2EDM at PSI	2026–2028	30,000,000	3.04%
JUNO	JUNO final precision	2028–2030	50,000,000	5.07%
CP_PHASE	DUNE + Hyper-K	2029–2033	40,000,000	4.05%
HIGGS_MTC	HL-LHC ATLAS+CMS	2030–2033	80,000,000	8.11%
STRONG_COUPLING	FCC-ee Z-pole	2040–2045+	100,000,000	10.13%

Remaining unminted supply (679M) is reserved for additional sectors added via governance and for platform extension.

4.3 Per-Milestone Distribution

Every milestone emission, regardless of outcome, distributes via MilestoneEmission.sol:

```
Example: JUNO sector resolves (50,000,000 ARTS minted)
  ├─ 25,000,000 (50%) → HypothesisEvolutionFund (research grants)
  ├─ 15,000,000 (30%) → Aerodrome LP              (protocol liquidity)
  ├─  7,500,000 (15%) → Staking rewards pool       (staker incentives)
  └─  2,500,000  (5%) → Burn                        (supply reduction)
```

4.4 Spiral Burn

PhiCoin implements a 0.618% (≈ $1/\varphi^2$) burn on every token transfer when total supply exceeds the floor of 9,200,000 ARTS. Addresses registered in `spiralBurnExempt` (staking contracts, liquidity pairs, critical infrastructure) are exempt from burn to prevent double-taxation during routine operations.

4.5 Staking

PhiStaking V3 offers three lock tiers with Fibonacci durations and golden-ratio multipliers:

TIER	LOCK DURATION	MULTIPLIER
0	5 days	1.0×
1	21 days	$\varphi \approx 1.618\times$
2	55 days	$\varphi^2 \approx 2.618\times$

Initial APY is 61.8% (= $1/\varphi$), decaying by a factor of $1/\varphi$ each weekly epoch. Staking rewards are paid from the staking rewards pool (funded by 15% of each milestone emission).

4.6 Fees

- — 1.18% on every PhiAMM swap, routed to protocol treasury
- — 9.02% of losing-pool payouts routed to the founder wallet as transparent on-chain royalty

Both rates are golden-ratio derived and fully disclosed in contract source code.

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5. Governance

5.1 Administration

The protocol is administered by a Gnosis Safe multisig at `0x75BA1367c9B2B750A1751Dd527902e0f1d67a8fb`. The Safe currently operates as 1-of-1 with the founder as sole owner; migration to a 2-of-3 multisig with additional ORCID-verified signers is planned.

5.2 Role-Based Access Control

Contract operations are gated by OpenZeppelin AccessControl roles:

ROLE	HELD BY	AUTHORITY
`DEFAULT_ADMIN_ROLE`	Safe multisig	Grant and revoke other roles
`UPGRADER_ROLE`	Safe multisig	Authorize UUPS upgrades (PhiCoin, PhiStaking V3, DiscoveryStaking)
`ORACLE_ROLE`	ScienceOracle contract	Call `resolveSector()` on ExperimentMilestoneProtocol
`ARBITER_ROLE`	ResearcherRegistry 2-of-3	Resolve oracle disputes

ROLE	HELD BY	AUTHORITY
`APPROVER_ROLE`	Safe multisig	Approve grants and bounties in HypothesisEvolutionFund
`MINTER_ROLE` (on PhiCoin)	MilestoneEmission + PhiStaking V3	Mint ARTS for emissions and staking rewards

5.3 Minting Authority

No role grants minting authority to any individual wallet. Minting requires the full contract chain:

```
ExperimentMilestoneProtocol → MilestoneEmission → PhiCoin.mintTo()
```

This chain requires a resolved sector, which requires oracle consensus (AI + 7-day challenge) or ORCID-verified arbitration. The founder cannot unilaterally mint ARTS.

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6. Experimental Roadmap

Token emission is driven by real-world experimental schedules.

YEAR	EVENT	SECTOR	EMISSION
2026–2028	n2EDM at PSI — final d_n result (θ_QCD test)	STRONG_CP	30M ARTS
2027	JUNO intermediate	JUNO (informational)	—

YEAR	EVENT	SECTOR	EMISSION
	data (precision ~1%)		
2028–2030	JUNO final $\sin^2\theta_{12}$ to 0.5% precision	JUNO	50M ARTS
2029–2033	DUNE Phase I δ_CP measurement	CP_PHASE	40M ARTS
2030–2033	HL-LHC ATLAS+CMS m_H to 21 MeV	HIGGS_MTC	80M ARTS
2032	Hyper-Kamiokande δ_CP cross-check	CP_PHASE	(completes sector)
2040–2045+	FCC-ee Z-pole α_s to 0.1% precision	STRONG_COUPLING	100M ARTS

Four of the five sectors are expected to resolve between 2026 and 2033.

7. Risk Disclosure

7.1 Scientific Risk

The Artosphere Hypothesis has an honestly assessed prior probability of 15–16% of being correct. There is approximately an 85% chance that one or more Tier 1 predictions will be refuted by future experiments.

The protocol is designed to continue functioning and creating value in the refutation scenario — refuted predictions trigger the same emission as confirmed ones, and the HypothesisEvolutionFund finances alternative research directions. However, long-term token value may be materially impacted by scientific outcomes.

7.2 Technical Risk

- All contracts are open-source under MIT license. A full third-party audit has not been commissioned at the time of writing. A competitive audit is planned for Q2–Q3 2026.
- The AI oracle may produce incorrect result submissions. The 7-day challenge window and human arbitration fallback mitigate but do not eliminate this risk.
- PhiCoin, PhiStaking V3, and DiscoveryStaking are UUPS upgradeable contracts. The `UPGRADER_ROLE` is held by the Safe multisig. Upgrades are not time-locked.
- The Safe currently operates 1-of-1. Loss or compromise of the key would result in loss of administrative control until migration to 2-of-3 is completed.

7.3 Regulatory Risk

PhiAMM is a spot automated market maker without leverage. Artosphere has not conducted an ICO, presale, or public sale. The protocol has not been tested in court and no legal opinion guarantees any specific regulatory classification.

7.4 No Investment Advice

This whitepaper is a technical and scientific description of the Artosphere protocol. It is not investment advice, solicitation, or an offer to sell securities. ARTS is a research instrument tied to experimental outcomes with a 15% prior probability of the underlying hypothesis being correct. Prospective holders should assume maximum loss and deploy only capital they can afford to write off.

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8. Deployed Contracts

All contracts are live on and verified on Basescan.

8.1 Token and Staking Layer

CONTRACT	ADDRESS
PhiCoin (ARTS)	`0x1C11133D4dDa9D85a6696B020b0c48e2c24Ed0bf`
PhiStaking V3	`0x5ba76643E3ef93Ab76Efc8e162594405A3c79f7B`
DiscoveryStaking	`0x3Fc4d3466743e0c068797D64A91EF7A8826a19e2`
PhiAMM	`0xf32c97846963c335eb78969c8c732945edc4e575`
NashFee	`0xb11e81168f97b6241cb037d9d02b282879ec3e52`
ResearcherRegistry	`0x295410735a0d9f68850a94b97a43fff7a5961cc9`
HypothesisEvolutionFund (seed)	`0x5c818B269A484321D650b526e4d47cF8D29dCF4B`

8.2 Governance

CONTRACT	ADDRESS
Gnosis Safe (multisig admin)	`0x75BA1367c9B2B750A1751Dd527902e0f1d67a8fb`

8.3 Experiment Milestone Protocol v3

The four v3 contracts (ExperimentMilestoneProtocol, ScienceOracle, MilestoneEmission, new HypothesisEvolutionFund) are written and compiling. Deployment addresses will be published in the authoritative `DEPLOYED_ADDRESSES.md` file upon launch.

8.4 Source Code

All Solidity contracts, Foundry scripts, and test suites are available at:

`github.com/fbsmna-coder/artosphere-contracts`

License: MIT

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9. References

9.1 Artosphere Scientific Papers (Zenodo)

1. Sapronov, F.B. Framework: 36 Standard Model Parameters from {M_Planck, φ}. Zenodo. DOI: 10.5281/zenodo.19484143
2. Sapronov, F.B. Master Action — The Artosphere Hypothesis. Zenodo. DOI: 10.5281/zenodo.19481141
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9.2 External Scientific References

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- Esteban, I., Gonzalez-Garcia, M.C., Maltoni, M. et al. NuFIT 5.2: Three-Flavor Global Fit to Neutrino Oscillation Data.
- Particle Data Group (2024). Review of Particle Physics. PRD 110, 030001.
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- n2EDM Collaboration (2021). Design of n2EDM. EPJ C 81, 512.
- HL-LHC ATLAS+CMS Physics Projections (2025). ATL-PHYS-PUB-2025-018.

9.3 Technical References

- OpenZeppelin Contracts v5.x — `github.com/OpenZeppelin/openzeppelin-contracts`
- Gnosis Safe — `github.com/safe-global/safe-smart-account`
- Chainlink Functions — `docs.chain.link/chainlink-functions`
- Foundry — `github.com/foundry-rs/foundry`
- Base (Coinbase L2) — `base.org`

The protocol mints ARTS when experiments publish results. Every result — confirmation or refutation — creates emission, liquidity, and research funding. The token exists to sustain a cycle of scientific progress, not to bet on any single outcome.

— F.B. Sapronov